LAB 8: Project Progress Report

Subject: System Simulation Lab / Capstone Project

Submission Date: 05 September 2026

Project Title: Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data

Challenge Reference: MathWorks Excellence in Innovation Challenge — Project #255

Institution: Vidyalankar Institute of Technology (VIT), Mumbai

Department: Department of Electronics and Computer Science Engineering

Team Members:

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1. Executive Summary & Project Status

Since the submission of the initial working prototype in LAB 7, the project has progressed into the **Advanced Optimization & Validation Phase**. All 5 Core Project Tasks are complete, and both optional Advanced Extensions have been implemented in the codebase.

Progress Summary Dashboard

Task / Module	Target Milestone	Status	% Completion
Core Task 1: Real-Time Map & Elevation Ingestion	Google Maps Directions & Elevation API integration	Completed	100%
Core Task 2: Driving Cycle Synthesis for Simulink	Time-speed synthesizer & `timeseries` export	Completed	100%
Core Task 3: Virtual Vehicle Composer BEV Modeling	Base workspace mapping & 10 Indian BEVs database	Completed	100%
Core Task 4: Powertrain Simulation & Energy Accounting	Longitudinal forces, 2D motor map, battery ECM	Completed	100%
Core Task 5: Visualization & Mapping Toolbox	Satellite/roadmap tiles, elevation & SOC plotting	Completed	100%
Advanced Task 1: Driving Mode Optimization	Eco, Normal, Performance mode trade-off engine	Completed	100%
Advanced Task 2: Highway Fast Charging Planner	Reserve violation detection, CC-CV taper model	Completed	100%

2. Implementation of Advanced Project Extensions

2.1 Advanced Extension 1: User-Informed Driving Modes (`optimizeDrivingMode.m`)

Three distinct driving profiles were calibrated to simulate behavioral impacts on powertrain efficiency:

1. **ECO Mode**:

- Acceleration limit: $a_{max} = 1.2\,m/s^2$

- Cruising speed cap: *v_{cap} = 85\,km/h*

- Regenerative braking prioritization: *100\%* regen capture before mechanical friction brake engagement.

- Auxiliary HVAC load: Eco compressor setting (*0.8\,kW*).

2. **NORMAL Mode**:

- Acceleration limit: *a_{max} = 1.8\,m/s^2*

- Cruising speed cap: *v_{cap} = 100\,km/h*

- Standard regen blending: *80\%* recovery efficiency.

- Auxiliary HVAC load: Standard comfort setting (*1.5\,kW*).

3. **PERFORMANCE Mode**:

- Acceleration limit: *a_{max} = 2.8\,m/s^2*

- Cruising speed cap: *v_{cap} = 120\,km/h* (expressway legal ceiling)

- Dynamic sport regen: *60\%* capture ratio with high wheel torque response.

- Auxiliary HVAC load: High cooling capacity (*2.2\,kW*).

Mode Comparison Results (Mumbai-Pune Corridor, Tata Nexon EV LR, 148 km):

Driving Mode	Travel Time	Net Energy	Avg. Efficiency	Final SOC	Trip Cost	Energy Savings vs Performance
ECO	*2h 14m*	*21.4\,kWh*	*14.4\,kWh/100km*	***47.4\%***	■\,*192.60*	***+27.2\% saved***
NORMAL	*2h 08m*	*25.1\,kWh*	*16.9\,kWh/100km*	***39.2\%***	■\,*225.90*	***+14.6\% saved***
PERFORMANCE	*1h 54m*	*29.4\,kWh*	*19.8\,kWh/100km*	***29.6\%***	■\,*264.60*	Baseline

> ***Notice: Key Finding:*** *Driving in Eco mode saves 8.0\,kWh of battery energy (equivalent to 45\,km of additional range) while only increasing travel time by 20 minutes across the 148 km trip.*

2.2 Advanced Extension 2: Highway Fast Charging Optimization (`optimizeCharging.m`)

When a planned route exceeds the vehicle's safe single-charge radius, the platform automatically plans charging stops:

1. **Reserve Violation Detection**:

Monitors instantaneous battery State of Charge. If *SOC(t) < 15\%*, a violation flag is triggered.

2. **Candidate Station Spatial Query**:

Scans the corridor for verified DC Fast Charging stations (Tata Power EZ Charge, Jio-bp pulse, Zeon Electric, etc.).

3. **Non-Linear CC-CV Taper Model**:

Models constant-current (CC) high-speed charging up to 80% SOC followed by constant-voltage (CV) thermal taper, preventing unrealistic linear charging estimations:

t_{charge} = \frac{E_{needed}}{P_{charger} \cdot \eta_{charge} \cdot \kappa_{taper}}

Where *\kappa_{taper} = 0.85* accounts for the slowing charge rate above 70% SOC.

4. **Session Cost Calculation**:

Computes electricity billing in INR based on commercial EV fast-charging tariffs (■\,*18.50/kWh*).

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3. Technical Challenges Overcome

Challenge	Impact	Engineering Solution Implemented
Noisy Raw Elevation Data	Step discontinuities from raw GPS/Google elevation caused unrealistic high-frequency grade spikes.	Implemented a spatial moving-average smoothing filter (`sgolay` / `smoothdata`) to produce realistic, smooth road slope angles (\theta).
API Rate Limits / Quotas	Network failures or missing API keys could crash simulations during offline grading.	Developed automated fallback to 6 high-resolution pre-compiled `.mat` Indian highway corridors.
Simulink Integration Friction	Manually importing custom cycles into Simulink was error-prone for evaluators.	Created `export_route_to_simulink.m` which automatically writes variables to base workspace and saves `route_drive_cycle.mat`.

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4. Work Remaining for Final Phase

1. **Master Test Suite Integration**: Assemble all unit and integration tests into a single master verification harness (`runAllTests.m`).
2. **Comprehensive Final Documentation**: Compile physical derivations into `ModelAssumptions.md` and `SystemArchitecture.md`.
3. **Public GitHub Repository Deployment**: Push codebase, configure clean README, and post official submission on MathWorks Challenge Hub Discussion #128.
4. **Final Presentation & Slide Deck**: Prepare slide materials for final project viva (LAB 9).